Math 252 [redacted] **Quiz 7** **Name:**

**1.** Circle **only one** integral technique that is most helpful for solving each of the following integrals and fill in the blank for the integral technique you choose. You do not need to evaluate the integrals.

1). $\displaystyle\int \frac{\sin x}{\cos^2 x}\,dx$

**A**. $u$-substitution with $u$= .

**B**. trigonometric substitution with $x$=

**C**. integration by parts with $u$= and $dv$=

2). $\displaystyle\int \sqrt{x}\ln x\;dx$

**A**. $u$-substitution with $u$= .

**B**. trigonometric substitution with $x$=

**C**. integration by parts with $u$= and $dv$=

3). $\displaystyle\int \frac{x^2}{\sqrt{9-x^2}}\,dx$

**A**. $u$-substitution with $u$= .

**B**. trigonometric substitution with $x$=

**C**. integration by parts with $u$= and $dv$=

**2.** Evaluate $\displaystyle\int \frac{3}{(x-1)(2x+1)}\,dx$.

# Formula Sheet

$$\sin^2 x + \cos^2 x = 1, \quad \sin^2 x = 1 - \cos^2 x, \quad \cos^2 x = 1 - \sin^2 x$$

$$\sec^2 x = \tan^2 x + 1, \quad \tan^2 x = \sec^2 x - 1$$

$$\sin^2 x = \frac{1 - \cos 2x}{2}, \quad \cos^2 x = \frac{1 + \cos 2x}{2}, \quad \sin 2x = 2 \sin x \cos x$$

$$\sin A \cos B = \frac{1}{2}[\sin(A - B) + \sin(A + B)]$$

$$\cos A \cos B = \frac{1}{2}[\cos(A + B) + \cos(A - B)]$$

$$\sin A \sin B = \frac{1}{2}[\cos(A - B) - \cos(A + B)]$$

$$\int \cot x \csc x \, dx = -\csc x + C, \qquad \int \csc^2 x \, dx = -\cot x + C$$

$$\int \tan x \, dx = \ln|\sec x| + C, \qquad \int \cot x \, dx = \ln|\sin x| + C$$

$$\int \sec x \, dx = \ln|\sec x + \tan x| + C, \qquad \int \csc x \, dx = \ln|\csc x - \cot x| + C$$

$$\int \frac{1}{1 + x^2} \, dx = \arctan x + C, \qquad \int \frac{1}{\sqrt{1 - x^2}} \, dx = \arcsin x + C$$

$$\int \frac{1}{a^2 + x^2} = \frac{1}{a} \arctan\left(\frac{x}{a}\right) + C, \qquad \int \frac{1}{\sqrt{a^2 - x^2}} = \arcsin\left(\frac{x}{a}\right) + C$$